

NSR/AS Lab 5 – Public Key Cryptography, Pattarapol Tantechasa, 103883220

Pattarapol Tantechasa
Student
Swinburne University of Technology
Melbourne, VIC, Australia
103883220@student.swin.edu.au

Abstract— This report introduces Public Key Cryptography, with both theoretical background and hands-on exercise. The first part outlined theory of Public Key Cryptography and RSA algorithm with related mathematical equations. Furthermore, steps to break the RSA algorithm in practical lab exercise to dive deeper into how cipher text decryption works using MATLAB. Results show all commands, and result of successfully decryption of provided cipher text.

Keywords—RSA Algorithm, Cryptography, Encryption, Decryption, Public Key, Private Key, Cipher Text, RSA Algorithm, Cryptography,

I. INTRODUCTION TO PUBLIC KEY CRYPTOGRAPHY

Public Key Cryptography is a method of encrypting information that uses two separate keys: a public key for encryption and a private key for decryption. It allows secure communication without needing to share a secret key in advance.

A. RSA Algorithm

1) Key Generation

$$n = p * q \quad (1)$$

$$x = (p - 1)*(q - 1) \quad (2)$$

Choosing two large prime numbers would be the first step of the RSA Algorithm, called it “p” and “q”. Multiply these two to get a number, “n” (1), which will be used in both public and private keys. The next step is to calculate how many numbers less than “n” are coprime to “n”, the result represented as “x” using equation (2). The next step is to choose small number that is coprime with value of “x”, it will represent as “e”.

$$d = (d * e) \bmod x \quad (3)$$

Finally, calculate and find the modular inverse of “e” using equation (3), the result represents as “d”. At this point, **public key** as “[e, n]” and **private key** as “[d, n]” had been successfully generated.

2) Encryption and Decryption

$$c = (m^e) \bmod n \quad (4)$$

$$m = (c^d) \bmod n \quad (5)$$

Encryption a message into cipher text using equation (4). Where “m” represents message and “c” represents cipher text, together with public key “[e, n]” will result as cipher text and the message had been successfully encrypted. Similarly, the decryption of cipher text is calculated with equation (5),

however with the value of private key “[d, n]”. At this point, the cipher text had been successfully decrypted.

II. BREAKING THE RSA ALGORITHM

The main goal of this lab is to decrypt a given cipher text with provided public key and function. No encryption is required. Therefore, the process of breaking the RSA algorithm in this lab focuses only on finding private key and using it to decrypt provided cipher text.

Command Window

```
>> % Set Up all Variables
c1 = [2377 400 2296 640 1142 1770 2014 540 1188
n1 = 2407;
e1 = 57;
```

Fig. 1. Set up command with provided public key and cipher text.

This command sets up a provided variables needed for RSA decryption. The cipher text represents with variable “c1”, public key represents with variables “n1” and “e1”.

Command Window

```
>> % Factor Equation: Get factor numbers and assign to p and q
factors1 = factor(n1);
p1 = factors1(1);
q1 = factors1(2);
```

Fig. 2. Command of Factor equation to find value of p and q.

In RSA, the product of two large prime numbers “p1” and “q1” is represented with “n1”. Therefore, to break the RSA encryption, the first step is to factor “n1”. MATLABs built in function “factor()” was used to find the prime factors of “n1”.

Command Window

```
>> % Compute for Private Key - d
x1 = (p1 - 1) * (q1 - 1);
for d1 = 1:x1
    if mod(d1 * e1, x1) == 1
        break
    end
end;
```

Fig. 3. Command to calculate private key.

This step computes the private key “d1”, first calculates the value of “x1” with equation (2). Then use that to perform calculations with equation (3). The for loop is used to tries every number from 1 up to the number “x”, checking if the result is equal to 1 or not. If it results in 1, break out of the loop and keep that value as the correct private key.

```

Command Window
>> % Decrypt ciphertext using Private Key with DecryptString function
message1 = decryptString(n1, d1, c1);

```

Fig. 4. Command of the cipher text decryption.

At this point, the program had all variables needed for the cipher text decryption. The decryption happens here where the decryption function had been called with parameters of “(n1, d1, c1)”, this satisfies the decryption equation (5). Finally, these are all steps and command needed for this lab to successfully decrypt the provided cipher text. Next is to repeat the process with the next provided cipher text and public key.

III. RESULTS

```

Command Window
>> % Patarapol Tantechasa, 103883220
% 1. First ciphertext to be decrypt
c1 = [2177 489 2796 648 1142 1778 2814 548 1188 1285 586 1421 1778 586 1142 2395 648 478 648 1285 2395 1778 586 1142 1778 1421 478 648 2814 2395 476 3838 478 1274]
% 2. Public Key - {n, e}
n1 = 2487;
e1 = 57;
% 5. Factor Equation: Get factor numbers and assign to p and q
factor1 = factor(n1);
p1 = factor1(1);
q1 = factor1(2);
% 7. Compute for Private Key - d
d1 = (p1 - 1) * (q1 - 1);
for d1 = 1:d1
    if mod(e1 * d1, d1) == 1
        break;
    end;
end;
% 8. Decrypt ciphertext using Private Key with DecryptString function
message1 = decryptString(n1, d1, c1);
% 11. Presentation text strings
disp('Message One:')
disp(message1)

```

Fig. 6. Screenshots of commands and message for cipher text 1.

This is a screenshot of all commands entered for the decryption of cipher text 1. It covered all steps mentioned in *Breaking the RSA Algorithm* section, as well as the decryption message.

Name	Value
c1	1x101 double
n1	1289
e1	57
factor1	[29,93]
message1	"When public interest in cryptography was just emerging in the late seventies and early eighties, the National Security Agency made several attempts to quash it."
p1	29
q1	83
d1	2796

Fig. 7. Screenshots of workspace for cipher text 1.

This is a screenshot of MATLAB workspace for the decryption of cipher text 1. It shows all variables name and values.

```

Command Window
>> % Patarapol Tantechasa, 103883220
% 2. Second ciphertext to be decrypt
c2 = [4514 5383 2971 7138 1332 2682 4485 6869 5363 1332 6869 5363 7138 1332 1889 2957 2682 5943 2971 4589 1332 1889 5561 2988]
% 4. Public Key - {n, e}
n2 = 7683;
e2 = 89;
% 5. Factor Equation: Get factor numbers and assign to p and q
factor2 = factor(n2);
p2 = factor2(1);
q2 = factor2(2);
% 8. Compute Private Key - d
d2 = (p2 - 1) * (q2 - 1);
for d2 = 1:d2
    if mod(e2 * d2, d2) == 1
        break;
    end;
end;
% 10. Decrypt ciphertext using Private Key with DecryptString function
message2 = decryptString(n2, d2, c2);
% 12. Presentation text strings
disp('Message Two:')
disp(message2)

```

Fig. 8. Screenshots of commands and message for cipher text 2.

This is a screenshot of all commands entered for the decryption of cipher text 2. It covered all steps mentioned in *Breaking the RSA Algorithm* section, as well as the decryption message.

Name	Value
c2	1x127 double
d2	3113
e2	89
factors2	[79,97]
message2	"This gave both the public practice of cryptography lots of unexpected publicity." Whitfield Diffie in "Applied Cryptography," (abridged)"
p2	79
q2	97
x2	7488

Fig. 9. Screenshots of workspace for cipher text 2.

This is a screenshot of MATLAB workspace for the decryption of cipher text 2. It shows all variables name and values.

IV. CONCLUSION

This lab introduced the RSA algorithm, covering its theoretical foundation and the key equations used in public key cryptography. The process of breaking RSA encryption is explored by recovering the private key from a given public key and cipher text, then used a provided decryption function with the calculated private key to decrypt the cipher text and retrieve the original message.

This exercise provided valuable insight into the mathematical foundation of RSA and demonstrated how its security depends on the difficulty of factoring large prime numbers.

In conclusion, this lab covered introduction to RSA algorithm, needed steps and equations to break RSA algorithm. As well as practical hands-on experience were conducted using MATLAB. All commands, theory behind it and results have been documented and explained in this report.

REFERENCES

[1] P. Branch. (2025). lab5 [PDF]. Available: https://swinburne.instructure.com/courses/66714/files/37999299/module_item_id=5005116